Dinghao Luo, PhD

PhD in Systems Neuroscience at Max Planck Florida Institute for Neuroscience

CONTACT

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EDUCATION

PhD, Neuroscience
2021-26 **Max Planck**

MS, Biological Sciences
2021-23 **Max Planck**

BA, Natural Sciences
2017-20 **Cambridge**, **1st**

RESEARCH METHODS

Electrophysiology
Two-photon imaging
Optogenetics
Behaviour training
Computational modelling

PROGRAMMING & DATA ANALYSIS

Python, **SQL**, **C++ (Arduino/Teensy)**
Git, **GitHub**
R, MATLAB
NWB/HDF5
CuPy

Statistical modelling
Machine learning
Image segmentation
Time-series analysis
Bayesian decoding
Permutation & bootstrap
Data validation
Multi-stream integration
GPU-accelerated computing

LANGUAGES

Mandarin (native)
Cantonese (native)
English (fluent)
German (intermediate)

AWARDS & HONOURS

Jardine Scholarship
Full competitive funding at University of Cambridge

IMPRS Fellowship
Ranked amongst the top incoming candidates

Pingshan Book Award
Chinese ed. of *Entangled Life*

Systems neuroscience researcher studying how locus coeruleus dopamine signals shape hippocampal dynamics and goal-directed behaviour. Combines causal experiments, multimodal neural recording, and computational modelling, with experience building reproducible software and analysis pipelines for large datasets.

EXPERIENCE

2021–present **PhD** in systems neuroscience
Max Planck Florida Institute for Neuroscience
- Co-conceived and led a multimodal project investigating how locus coeruleus dopamine alters hippocampal population dynamics and goal-directed behaviour.
- Designed experiments combining silicon-probe electrophysiology, two-photon imaging, optogenetics, local pharmacology, and virtual-reality behaviour.
- Built end-to-end Python pipelines for 273 two- sessions across 28 animals and 106 electrophysiology sessions across 19 animals, including validation, synchronisation, segmentation, population analysis, and visualisation.
- Applied GLMs, Bayesian decoding, permutation inference, dimensionality reduction, and computational modelling to test competing mechanisms.
- Built MATLAB/Psychtoolbox and Arduino/Teensy acquisition systems and resolved acquisition noise, dropped-frame, and timestamp failures.
- Integrated experimental and computational results into a first-author manuscript under review at *Neuron*. *bioRxiv*: https://doi.org/10.64898/2026.09.02.748730.

2019–2020 **Undergraduate Researcher** in neuroscience
University of Cambridge
- Designed and analysed whole-cell patch-clamp experiments on dopamine-dependent hippocampal synaptic plasticity; thesis awarded first-class marks.

2018 **Genetics Society Studentship** in neuropathology
University College London
- Analysed cathepsin B activity and protein abundance in Down syndrome-Alzheimer's disease mouse models; co-authored a 2023 peer-reviewed paper.

SELECTED SOFTWARE & DATA PROJECTS

FibreSight | Image segmentation, computer vision & scientific software [GitHub]
- Designed, trained, and evaluated PyTorch U-Net and attention U-Net segmentation models; created 1,296 curated ROIs across 53 sessions, and achieved 0.93 recall and 0.83 F_2 on nine held-out sessions with the released model, reducing manual curation from days to hours.
- Built and released a TIFF-to-NWB workflow covering image registration, per-frame quality checks, ROI segmentation, human correction, fluorescence extraction, and dF/F calculation, with source hashes, checkpoint identities, run parameters, and named analysis runs for provenance.
- Released the system as an open-source Python/PyQt5 workbench with automated tests, continuous integration, and documentation.

Multimodal neural-data pipeline | Scientific programming, statistics & GPU computing [GitHub]
- Built and maintained an end-to-end Python pipeline for electrophysiology, two-photon imaging, and behavioural data, including ingestion, cross-stream synchronisation, dropped-frame detection, data QC, cached intermediate outputs, documented analysis stages, and reproducible figures.
- Used CuPy, vectorisation, multiprocessing, and memory optimisation to reduce a projected weeks-long calculation to hours.

PUBLICATIONS & MANUSCRIPTS

- **Luo, D.**, *et al.* Spatially confined dopamine from locus coeruleus axons selectively shapes hippocampal dynamics and action timing. Under review at Neuron (2026). https://doi.org/10.64898/2026.09.02.748730.
- Wu, Y., et al., including **Luo, D.** Cathepsin B abundance, activity and microglial localisation in Alzheimer's disease-Down syndrome and early onset Alzheimer's disease: the role of elevated cystatin B. *Acta Neuropathologica Communications* 11, 132 (2023). https://doi.org/10.1186/s40478-023-01632-8.

SELECTED PRESENTATIONS

- Locus coeruleus dopamine rapidly modulates CA1 dynamics to sharpen goal-directed navigation. **COSYNE**, Lisbon, Portugal, 12-17 March 2026.
- Locus coeruleus dopaminergic modulation of CA1 shapes behavioural timescale dynamics for time integration. **Lake Conference**, Seattle, USA, 28 September-2 October 2025.
- Locus coeruleus regulates hippocampus-dependent integration at single-trial level. Memory Formation Forum, Max Planck Florida Institute for Neuroscience, Jupiter, USA, 18 September 2024.
- An LC-CA1 circuit for initiating path integration. Sunposium, West Palm Beach, USA, 6 March 2024.
- Mapping the causal gene for cathepsin B activity deficits in DS-AD. Genetics Society Summer Workshop, Edinburgh, UK, 27-30 August 2018.

INDEPENDENT COMPUTATIONAL & DIGITAL PROJECTS

The Zone | Digital humanities, literary analysis & interactive data visualisation [The Zone]

- Built a computational literary-analysis pipeline and interactive Astro/D3 website for Thomas Pynchon's *Gravity's Rainbow* (1973).
- Using NLP methods, curated 138 entities and 247 aliases, then resolved 5,087 mentions using deterministic context-aware rules across the entire novel.
- Developing interactive character-network, episode-density, and lexical views in TypeScript and D3, with search, filtering, responsive layouts, and provenance-preserving data.

Kairos | Machine learning & bibliometrics [GitHub]

- Built a Python ML pipeline with OpenAlex data to ask whether an academic field's state at a given year can predict whether scientific breakthroughs are imminent, using manually reviewed labels, temporally lagged features, and expanding-window validation.
- Compared field-state predictors with flexible year and publication-volume baselines; the current analysis does not establish predictive information beyond those simpler baselines.

SCIENCE COMMUNICATION, TEACHING & MENTORING

- **Published translator**:
  - Translated Merlin Sheldrake's Entangled Life (2020) into Chinese, published by Houlang Books in 2025; recipient of the 2025 Pingshan Natural History Museum Book Award.
  - Translated A. David Redish's *The Mind Within the Brain* (2013) into Chinese; scheduled for publication in late 2026.
- **Editor** and **science writer** at *Scientific American China* (2020–2021); edited neuroscience and biomedical features and conducted expert interviews with prominent scientists; published a print feature on the Templeton-funded Accelerating Research on Consciousness initiative.
- **Writer**, **editor**, and **section lead** at Neu-Reality (2017–); wrote and edited neuroscience material and coordinated a 35-person contributor group.e
- **Mentored** post-baccalaureate researchers, technicians, and incoming laboratory members in Python, Git, analysis workflows, debugging, and experimental protocols.
- **Tutored** high-school students in biology, statistics, psychology, and computer science.

TRAINING CERTIFICATES

- SQL Associate, DataCamp, certified July 2026.
- CITI Responsible Conduct of Research, Florida Atlantic University, completed November 2023.